A

Mini Project Report

on

Campus Trace: Lost and Found Items within Campus

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science Engineering (Data Science)

by

Parth Talwande	**24107003**
Akshat Ukheda	**24107026**
Amay Turbhekar	**24107020**
Anmol Shelar	**24107128**

Under the guidance of

Dr. Avinash Shrivas

DEPARTMENT OF COMPUTER SCIENCE ENGINEERING (DATA SCIENCE)

A.P. SHAH INSTITUTE OF TECHNOLOGY

G.B. Road, Kasarvadavali, Thane (W)-400615

UNIVERSITY OF MUMBAI

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CERTIFICATE

This to certify that the Mini Project report on Campus Trace: Lost and Found Items has been submitted by Parth Talwande (24107003), Amay Turbhekar (24107020), Akshat Ukheda (24107026) and Anmol Sheklar (24107128) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Dr. Avinash Shrivas
Guide

Dr. Pravin Adivarekar
HOD, CSE(Data Science)

Dr. Uttam D. Kolekar
Principal

External Examiner:
1.

Internal Examiner:
1.

Place: A. P. Shah Institute of Technology, Thane
Date:

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Chapter 1

Introduction

In every educational institution, loss of personal belongings is a common problem. Students and staff frequently misplace items such as ID cards, wallets, keys, earphones, and laptops across campus areas including classrooms, libraries, cafeterias, and common areas.

Campus Trace is a web-based application designed to solve this problem by providing a centralized, organized, and easily accessible digital platform for managing lost and found items within a campus environment. The application allows any registered user student or staff to post a report of a lost or found item, including photographs and location details, and enables direct communication between the person who lost an item and the person who found it. The entire process that previously relied on chance and informal communication is streamlined into a structured, searchable, and responsive system.

1.1. Purpose:

- To present the design, development, and implementation of a web-based Lost and Found management platform for campus environments.
- To provide students and staff with a structured digital alternative to informal methods such as notice boards and WhatsApp groups.
- The target audience includes college students, faculty members, and administrative staff operating within a campus setting.
- To enable quick and direct communication between the finder of an item and its original owner, eliminating reliance on third-party platforms.

To create a scalable system that can be extended to multiple campuses or institutions in future phases.

1.2. Problem Statement:

Students and staff at college campuses routinely lose personal belongings such as ID cards, wallets, keys, earphones, and stationery items within the campus premises. At present, there is no dedicated centralized system to report or track lost and found items within the campus. The existing approaches such as word of mouth, WhatsApp groups, and physical notice boards are informal, unreliable, and do not guarantee that the message will reach the right person.

1.3. Objectives

- To design and implement the user interface for a campus lost and found system. The UI will be built using React and Tailwind CSS, ensuring a clean, and intuitive experience for all campus users across devices.
- To create a system that allows users to report lost and found items. Users will be able to submit detailed reports including item name, description, category, making it easier to identify and match items.
- To implement a messaging system for communication between users. A built-in contact feature will allow the finder and the owner to communicate directly within the platform, eliminating the need for any external communication tools.
- To implement search and filter functionality for items. Users will be able to search for items using keywords and apply filters based on category, helping them quickly narrow down and locate relevant listings.

1.4. Scope:

- Can be applied in college and university campuses to digitally manage and track lost and found items within the campus environment.
- Can be useful to students, teachers, and campus staff to quickly report lost belongings and help others recover their items

CHAPTER 2

PROPOSED SYSTEM

Campus Trace is a mobile-responsive web application that provides a centralized digital platform for managing lost and found items within a campus. The system allows users to post reports of lost or found items, upload photos, search through existing reports, and communicate directly with other users to facilitate the return of belongings. The application is built using modern web technologies and follows a component-based architecture for maintainability and scalability.

2.1. Features and Functionality

- User Registration and Login Users can create an account using their campus credentials and securely log in to access the system. This ensures that only verified campus members can post or browse reports, maintaining accountability within the platform.
- Report Lost Item Users can post a detailed report of a lost item including the item name, category, description, building and floor location, and an uploaded photograph. This makes it easier for finders to identify and match the item.
- Report Found Item Users who find an item on campus can submit a found report with a photo, location, and description. The report is immediately visible on the home feed, allowing the original owner to identify their belonging and get in touch.
- Search and Filter Items Users can search all reported items using keywords and filter by type (Lost or Found) or category (Electronics, Books, IDs, Accessories, Clothing), making it quick to find a specific item in the feed.
- Contact Owner/Finder The in-app messaging system allows direct, private communication between the finder and the owner without sharing personal contact details, making the item return process smooth and straightforward.

CHAPTER 3

Project Outcomes

The development of the Campus Trace application is expected to produce the following outcomes:

Centralized Lost and Found System:

The project creates a single digital platform where all lost and found items on campus can be reported, browsed, and tracked. This eliminates the fragmented approach of using notice boards or word of mouth and provides a structured record of all reports.

Faster Item Recovery:

With quick reporting, photo upload, and keyword search features, the chances of recovering lost items are significantly higher. Users can immediately post a report or search for their lost item, reducing the time taken to reconnect items with their owners.

Improved Communication:

The in-app messaging system allows direct, real-time communication between the person who found the item and the original owner. This removes the need for third-party communication and makes the handover process more direct and reliable.

Reduced Manual Effort:

Instead of relying on physical notice boards, social media posts, or verbal announcements, the system digitizes the entire lost and found process. This reduces the manual effort involved for both users and campus administration.

CHAPTER 4

Software Requirements

Operating System

- Android OS (for mobile deployment)
- Windows/macOS (development environment)

Development Tools

- Flutter SDK
- Visual Studio Code / Android Studio
- Node.js Runtime Environment
- Postman (for API testing)

Programming Languages

- Dart (primary language for frontend/app development)
- JavaScript (backend development with Node.js)

Libraries & Packages Used

- express → Used for building the backend server and handling API routes
- mongoose → Used for connecting and interacting with MongoDB database
- jsonwebtoken (JWT) → Used for secure user authentication and session management
- bcrypt → Used for hashing and securing user passwords
- dotenv → Used for managing environment variables securely
- cors → Used for enabling cross-origin requests between frontend and backend

- http (Flutter) → Used for making REST API calls from the Flutter app to the backend
- shared_preferences (Flutter) → Used for storing user session data locally on the device
- flutter_local_notifications (Flutter) → Used for sending in-app alerts and notifications to users

Database

- MongoDB (cloud-hosted via MongoDB Atlas)
- Mongoose ODM for schema definition and database queries

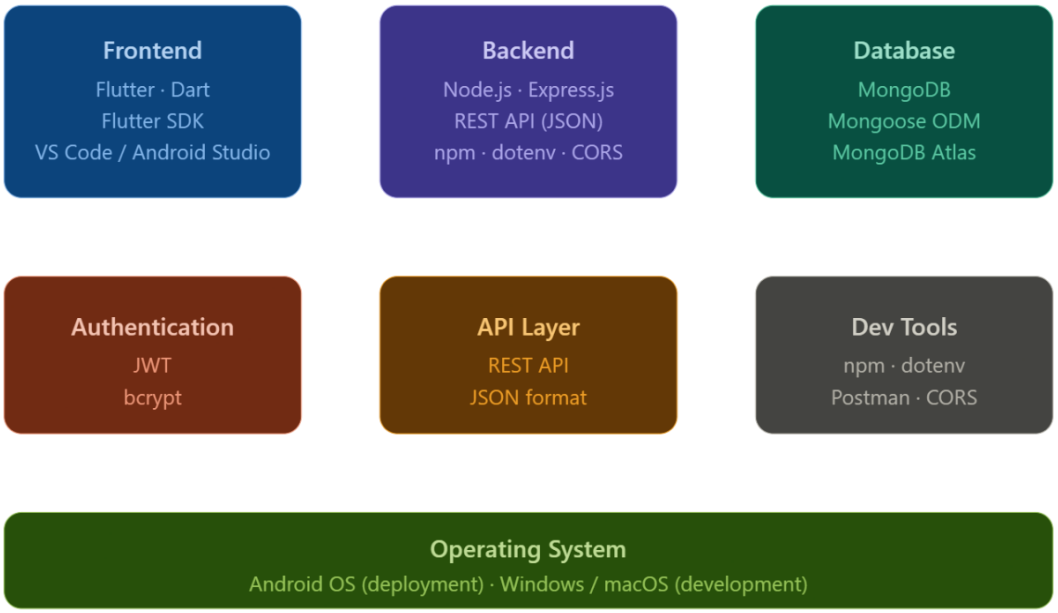


Figure 4: Technology Stack

CHAPTER 5

Project Design

Frontend:

- React 18 + TypeScript (UI framework and language)
- Vite (build and development server)
- Frontend communicates via → REST API (planned for future integration)

Backend :

- Node.js, Express.js (server framework)
- Authentication: → JWT (JSON Web Tokens) for secure user login and session management
- API: → REST API with JSON for communication between frontend and backend
- Tools: → npm, dotenv, CORS

Database :

- MongoDB, Mongoose → NoSQL cloud database for storing item reports, user data, and messages
- Storage: → Firebase Storage / Cloudinary for storing uploaded item photographs

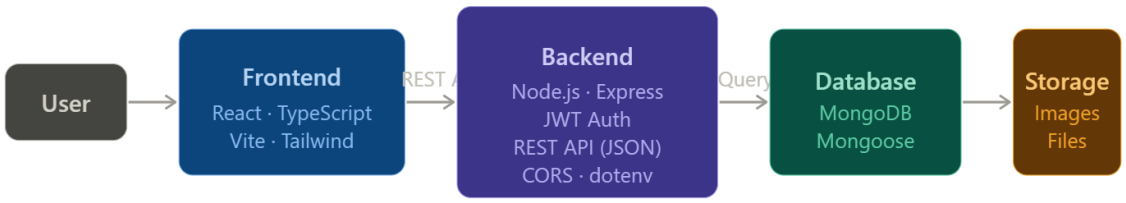


Figure 5.1: Dataflow

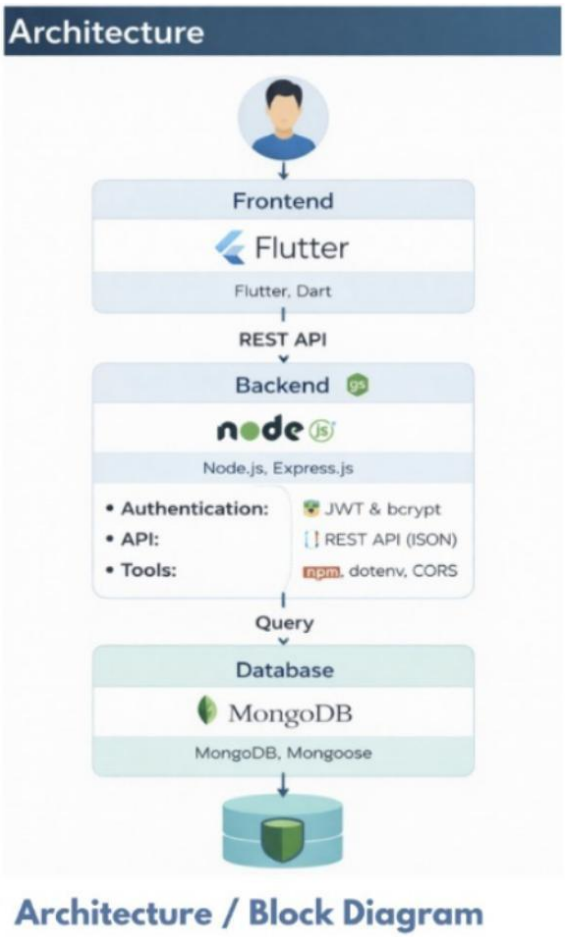
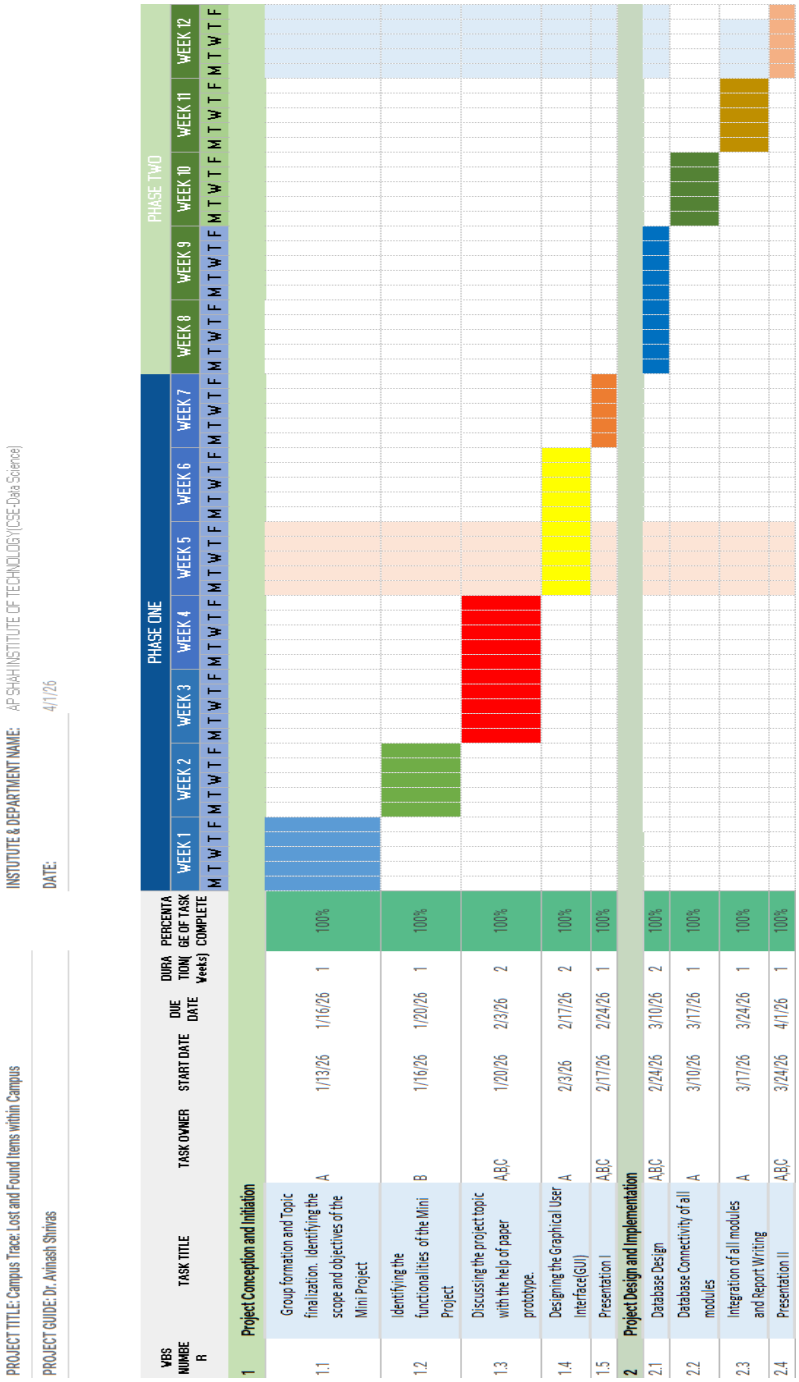


Fig5.2: Architecture

CHAPTER 6

Project Scheduling



CHAPTER 7

Results:

The Campus Trace application was successfully developed and tested as a fully functional lost and found platform for campus use. Users could register, log in, and access all features through a clean and responsive interface built with React 18, TypeScript, and Tailwind CSS with shadcn/ui components. The platform performed consistently across different screen sizes, providing a smooth user experience throughout.

The lost and found reporting modules worked as intended, allowing users to post item details including name, description, category, and last seen location. Found item reports supported image uploads alongside descriptions, which improved the accuracy of matching items with their owners. All entries were instantly stored in the centralized database and made visible to every registered user on the platform.

The search and filter functionality allowed users to quickly browse through reported items using keywords, categories, or location filters, significantly reducing the time and effort needed to find a match. React Router DOM ensured seamless navigation between pages, while Sonner-based toast notifications kept users informed of key actions in real time.

The in-app contact feature enabled direct communication between the finder and the owner without the need for any external platform, making the item return process straightforward and efficient. Overall, the application successfully replaced traditional manual methods such as notice boards and word-of-mouth communication with a fast, organized, and accessible digital solution, meeting all the objectives defined at the start of the project.

CHAPTER 8

CONCLUSION:

Campus Trace was developed as a solution to a common yet largely unaddressed problem faced by students and staff in college environments — the loss of personal belongings with no efficient means of recovery. The project successfully achieved its core objectives by providing a centralized digital platform for reporting, searching, and recovering lost and found items within the campus.

The application was built using a modern technology stack React 18, TypeScript, Vite, and Tailwind CSS ensuring a fast, scalable, and user-friendly experience. Features such as user, item reporting, category-based search, and direct communication between users significantly reduce the time and effort involved in recovering lost items.

Through this project, the team gained hands-on experience in full-stack web development, UI/UX design principles, state management using React Context API, and component-based architecture. The development process also enhanced our understanding of real-world problem solving through software engineering practices.

Overall, Campus Trace stands as a practical and impactful mini project that addresses a real need making campus life a little less stressful for everyone who has ever lost something and had no easy way to find it back.

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